

BillieBot — Hardware Build & Test Manual

Step-by-step assembly instructions with complete connection tables (interface, connector, and signal for every link), subsystem wiring schematics, and staged component/system test procedures. Rev A — MVP configuration.

1 · How to use this manual

Work the stages of Section 11 in order; each stage tells you which connections to make (by their connection IDs, e.g. **P6**, **D3**), and ends with a **checkpoint** you must pass before moving on. Sections 5–8 are the wiring reference: one section per subsystem, each with a connection table and a schematic. Every connection has an ID — **P-numbers are power, D-numbers are data/signal** — and Appendix A collects all of them into a single build checklist. Connections into/out of the Arduino Nano's individual pins are intentionally described only at the functional level (e.g. “quadrature pulses to the base controller”), since you already have the pin map; everything else is specified down to the physical connector.

Software checks reference the launch ladder (rungs 01–14) and test scripts (TC-xxx) from the BillieBot software package; run them from **billiebot_bringup/** and **billiebot_tests/scripts/**.

2 · Safety rules (apply to every stage)

- **LiPo:** charge only with a balance charger, inside a LiPo bag, attended. Storage-charge (~3.8 V/cell) when not in use. Retire the pack if any cell falls below 3.3 V or it puffs.
- **Wheels-up rule:** the first test of anything that can move happens with the robot on a stand, wheels off the ground.
- **Bench PSU first:** first power-up of any new wiring uses a current-limited bench supply (start at 0.5 A limit), never the LiPo.
- **Master switch** mounted on the top deck, reachable in one motion, before the robot ever drives on the floor.
- **One change at a time.** When something breaks, the last thing you touched is the suspect. Keep a build log (date, action, measurement).

3 · Tools and consumables

Tools: soldering iron + solder + flux, heat-shrink assortment, wire strippers/crimpers, multimeter, screwdriver set, zip ties, M2.5/M3 standoffs and screws, double-sided foam tape, hot glue.

Consumables to buy (small items not in your parts list): XT60 connector pairs; 16–18 AWG silicone wire (power) and 22–24 AWG (signal); Dupont jumper assortment; spade/ring terminals for the terminal strips; inline fuse holders matching your fuse kit; LiPo-safe bag + balance charger; a 12 V bench supply or current-limited PSU; one $\geq 1000 \mu\text{F}$ / 25 V electrolytic capacitor; one 10 k Ω + one 3.3 k Ω resistor (battery-sense divider); a 22↔15-pin FFC adapter cable if you will use the Pi Camera v2 (the Pi 5 uses 22-pin mini FFC connectors); optionally an 850 nm IR LED module + logic-level MOSFET for NoIR night vision.

4 · Component roles at a glance

Component	Subsystem	Role
Jetson Orin Nano Super (+NVMe)	Compute	Primary autonomy computer: SLAM, Nav2, mission behavior tree, OAK-D pipeline, base bridge
Raspberry Pi 5 (4 GB) + USB hub	Compute	Sensing companion: audio classifier + DoA, thermal, NoIR camera, state fusion, SQLite event log, daily report + web server
Arduino Nano V3	Drive	Base controller: encoder counting, PID wheel-velocity loop, L298N PWM, BNO055 host, battery ADC, heartbeat motor cutoff

Component	Subsystem	Role
RPLidar A1	Perception	2-D laser scan for SLAM, localization, costmaps
OAK-D Lite	Perception	Dog detection: on-device YOLOv8n + stereo depth → 3-D dog position
Pi Camera 3 NoIR / Pi Camera v2	Perception	Low-light imagery (needs IR illuminator) / optional spare
MLX90640 thermal (55°)	Perception	Warm-body (30–40 °C) blob detection in darkness, sleep confirmation
ReSpeaker XVF3800	Audio	4-mic array: bark/whine detection input + direction of arrival
MAX98357A + 3 W speaker	Audio	Status chimes (MVP); Speak action reserved for Behavior AI
SPH0645 I²S mic	Audio	Spare/backup microphone (unused in MVP)
2× JGA25-371 + 68 mm wheels + caster	Drive	Differential drive, ~0.46 m/s max
L298N dual H-bridge	Drive	Motor power stage (12 V)
BNO055 + BMP280 (Gravity 10-DOF)	Drive	Fused orientation for odometry EKF
Venom 3S LiPo 11.1 V 4000 mAh	Power	Main battery (44 Wh; ~60–75 min patrol)
Fuses, switch, terminal strips	Power	Protected star-topology distribution bus
GeeekPi PD board	Power	12 V bus → USB-C PD 5 V/5 A for the Pi 5
DFRobot GS2678 buck	Power	12 V → 5 V accessory rail (amp, IR LED)
Pololu D24V90F5	Power	RESERVED 5 V/9 A rail for future treat dispenser
GL-SFT1200 router	Network	Dedicated robot WLAN (stays in the apartment)

5 · Power subsystem — wiring and instructions

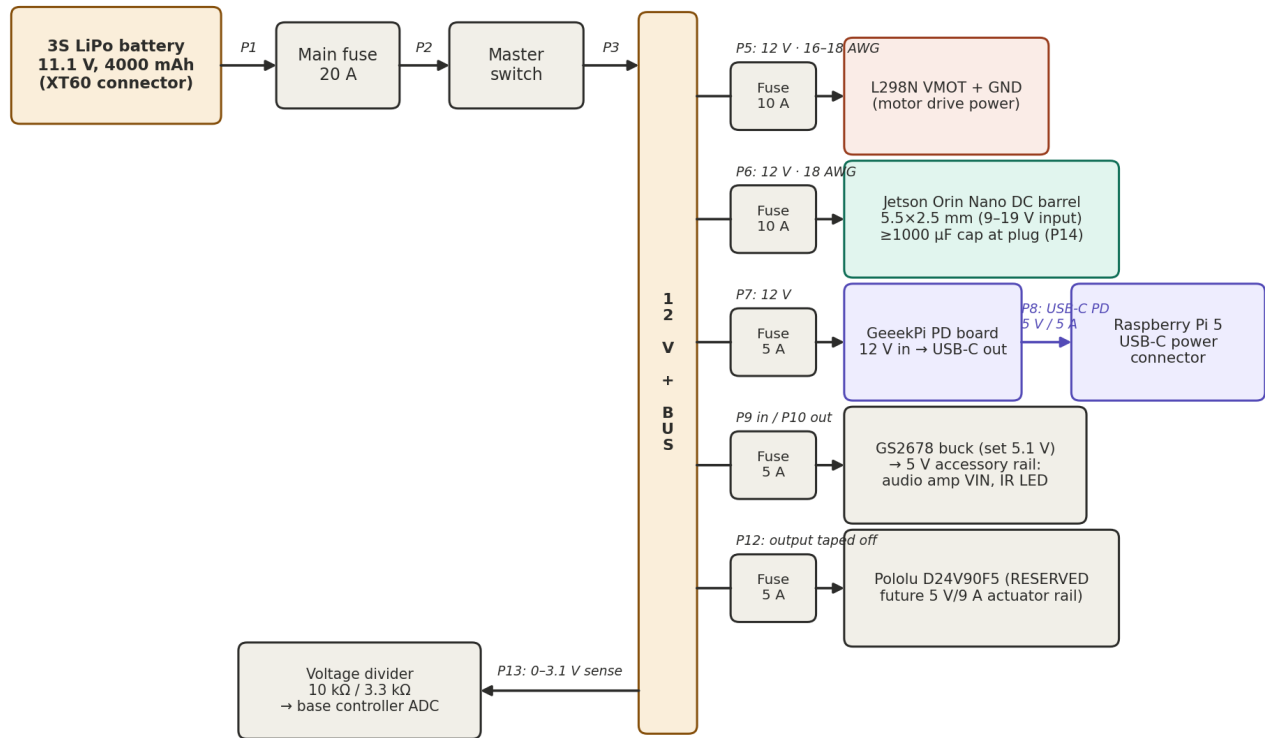
Build the entire power tree on the bench and verify every rail *unloaded* before any computer or sensor is connected. Solder the battery chain (P1-P4), land the five fused branches on the terminal strips but leave the load ends disconnected, set the GS2678 buck to 5.10 V under no load, then verify the table's voltages with the bench PSU at 12.0 V before repeating once, briefly, on the LiPo.

Table 5-1 · Power connections

ID	From (connector)	To (connector)	Interface	Signal carried	Electrical / cable notes
P1	Battery XT60 (+)	Main fuse holder	XT60 + solder	Raw battery power	9.9–12.6 V; 16 AWG; fuse 20 A
P2	Main fuse	Master switch	Spade/solder	Switched 12 V feed	16 AWG
P3	Master switch	+12 V bus terminal strip	Ring terminal	Bus supply	16 AWG; keep run short
P4	Battery XT60 (–)	GND bus terminal strip	Ring terminal	Ground reference	16 AWG; single star-ground point
P5	+12 V bus via 10 A fuse	L298N VMOT + GND screw terminals	Screw terminals	Motor drive power	Peaks ~4 A on stall (both motors); 16–18 AWG
P6	+12 V bus via 10 A fuse	Jetson DC barrel jack 5.5×2.5 mm, center-positive	Barrel plug	Compute power	Jetson accepts 9–19 V, so 3S direct is fine; ~1–2 A typical; 18 AWG
P7	+12 V bus via 5 A fuse	GeeekPi PD board VIN	Screw terminals	PD-converter input	Verify the board's printed input range covers 9.9–12.6 V
P8	GeeekPi PD board USB-C out	Pi 5 USB-C power connector	USB-C	5 V / 5 A USB-PD (negotiated)	Use the board's supplied/short USB-C cable
P9	+12 V bus via 5 A fuse	GS2678 buck IN	Screw terminals	Buck input	
P10	GS2678 buck OUT	5 V accessory rail (terminal strip)	Screw terminals	Regulated 5.1 V rail	Set to 5.10 V unloaded; ≤5 A
P11	5 V accessory rail	MAX98357A VIN/GND; IR LED module (via MOSFET)	Dupont / solder	Accessory power	22 AWG; amp draws <1 A peak
P12	+12 V bus (fused)	Pololu D24V90F5 VIN	Solder pads	RESERVED future rail	Output taped off until v2 treat dispenser
P13	+12 V bus	10 kΩ / 3.3 kΩ divider → base-controller analog input	22 AWG signal wire	Battery-sense analog	0–3.13 V over the pack's range (3.13 V at 12.6 V)
P14	Across Jetson barrel feed (P6)	—	Solder, mind polarity	Bulk ride-through capacitance	≥1000 μF / 25 V electrolytic at the plug end

Rules: every fuse sits at the *bus* end of its branch; motor (P5) and compute (P6–P8) wiring are separate home-runs to the strips — never daisy-chained; all grounds meet only at the GND strip. Do not connect the L298N's onboard “5 V” output pin to any of your rails (leave its 5 V-EN jumper ON so it feeds its own logic).

Figure 1 — Power distribution (connections P1-P14)



Ground: battery (-) to a GND terminal strip; every load returns to that strip (star ground). All fuses sit at the bus end of each branch.
Wire gauge: 16 AWG battery and motor runs, 18 AWG compute feeds, 22 AWG sense/signal.

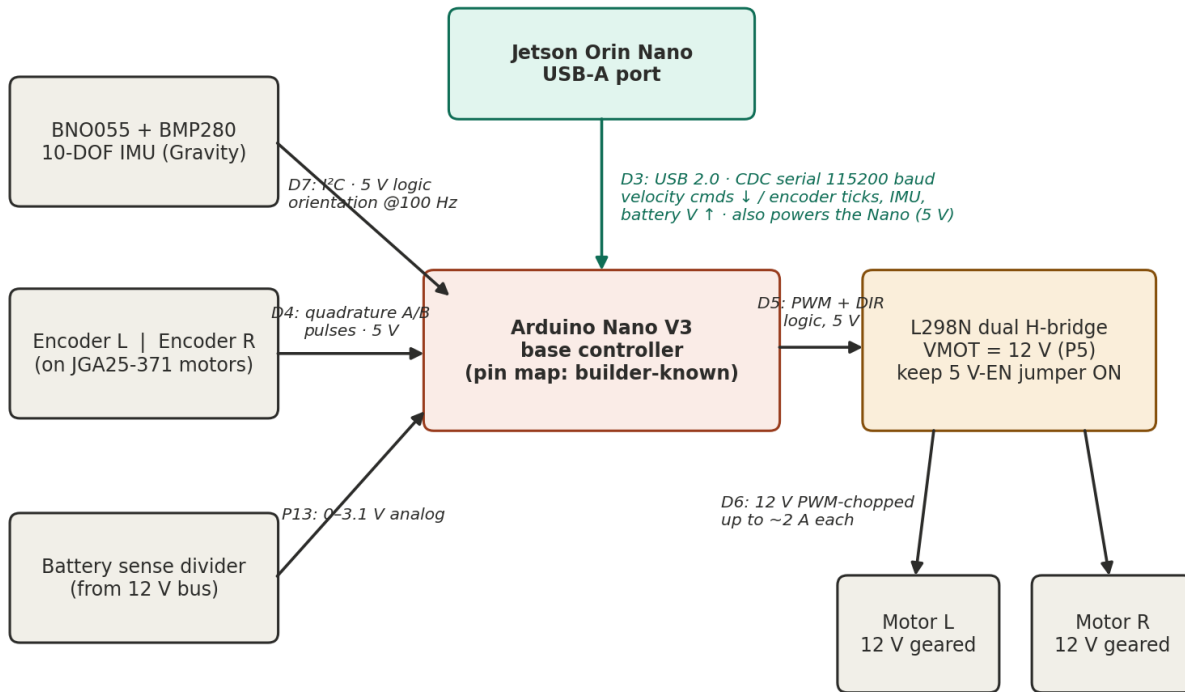
6 · Drive / base subsystem — wiring and instructions

The Arduino Nano is the base controller; its pin assignments are yours (per the reference firmware), so this section specifies each link functionally. Wire the drive electronics on the bench per Table 6-1, flash the reference firmware, and run the Stage 2 tests before mounting anything on the chassis. **Power the Nano from the Jetson's USB cable only (D3)** — do not also feed its 5 V pin from the accessory rail; a dual supply back-feeds the USB port.

Table 6-1 · Drive connections

ID	From (connector)	To (connector)	Interface	Signal carried	Electrical / cable notes
D3	Arduino Nano mini-USB	Jetson USB-A port	USB 2.0 cable	CDC serial 115200 8N1: wheel-velocity setpoints down; encoder ticks, IMU orientation, battery volts up (~50 Hz frames). Also 5 V power to the Nano	The Jetson-side node is the ros2 base bridge; heartbeat embedded in the command stream
D4	Encoder L and R (motor pigtails)	Base controller digital inputs	Dupont / solder	Quadrature A/B pulses, 5 V logic	Twist each A/B/5V/GND bundle; 22–24 AWG; route away from motor leads
D5	Base controller PWM/DIR outputs	L298N IN1–IN4 + ENA/ENB header	Dupont	5 V logic PWM (~1–20 kHz) + direction	Remove the ENA/ENB jumpers so PWM controls speed
D6	L298N OUT1/2 and OUT3/4	Motor L and Motor R terminals	Screw terminals	12 V PWM-chopped drive	≤2 A per motor; 18 AWG; if a wheel runs backward, swap that OUT pair
D7	BNO055 Gravity board SDA/SCL/VCC/GND	Base controller I ² C + 5 V	Dupont	I ² C orientation data (quaternion, gyro, accel) @100 Hz	Gravity board is 5 V-tolerant; mount near the robot's rotation center, away from motor magnets
P5	(power — see Table 5-1)	L298N VMOT			
P13	(power — see Table 5-1)	Base controller ADC			

Figure 2 — Drive / base subsystem (connections D3-D7, P5, P13)



Power the Nano from the Jetson USB cable only — do NOT also feed its 5 V pin from the accessory rail (no dual supply).
Twist each encoder A/B/5 V/GND bundle; route away from motor leads. If a wheel spins backward later, swap that motor's OUT pair at the L298N.

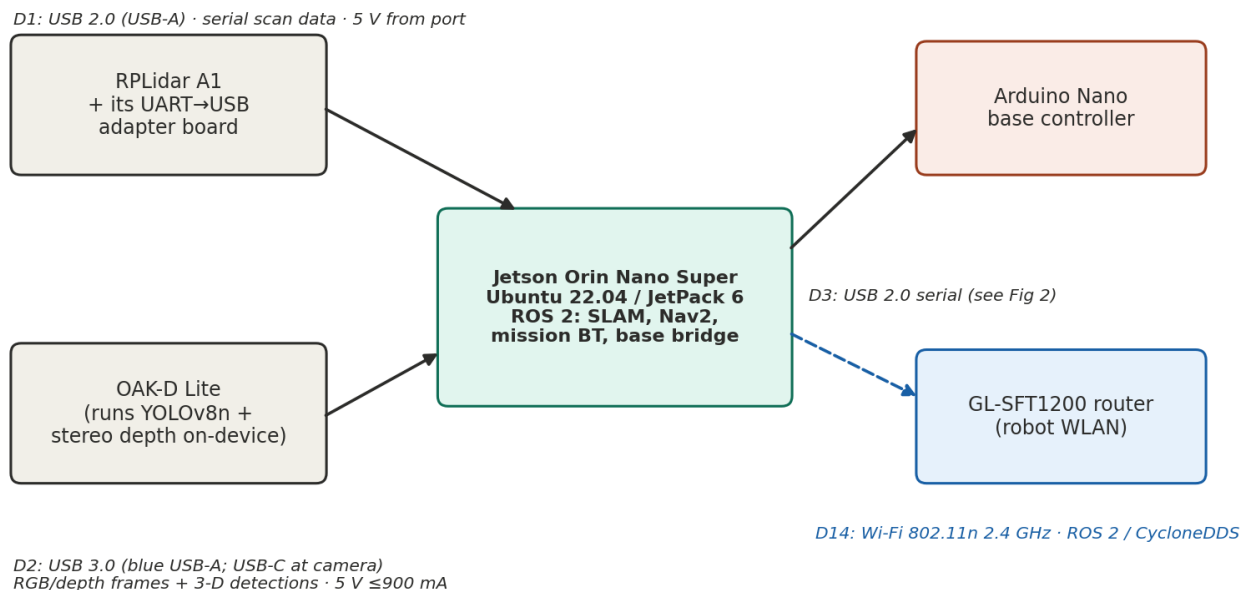
7 · Jetson data connections

Everything in the navigation-critical loop plugs into the Jetson so no Wi-Fi hop sits between perception and motor commands.

Table 7-1 · Jetson connections

ID	From (connector)	To (connector)	Interface	Signal carried	Electrical / cable notes
D1	RPLidar A1 JST pigtail → its UART-USB adapter board	Jetson USB-A (2.0)	USB 2.0	Serial laser-scan data (range/angle samples, ~8 kHz) + lidar motor control	Adapter draws ~450 mA of 5 V from the port
D2	OAK-D Lite USB-C	Jetson USB-A 3.0 (blue)	USB 3.0 SuperSpeed	RGB + stereo/depth streams and on-device YOLO detection results	≤900 mA of port 5 V — borderline; use a powered hub or Y-cable if it resets
D3	Arduino Nano	Jetson USB-A	USB 2.0	Base-controller serial (see Table 6-1)	
D14	Jetson Wi-Fi	GL-SFT1200 router	Wi-Fi 802.11n 2.4 GHz, WPA2	ROS 2 / CycloneDDS topics, SSH	Static DHCP lease (e.g. 192.168.8.10)
P6	(power — see Table 5-1)	DC barrel jack			

Figure 3 — Jetson Orin Nano data connections (D1-D3, D14)



If the OAK-D brownouts the port (random disconnects), insert a powered USB hub or a Y-cable fed from the 5 V accessory rail.

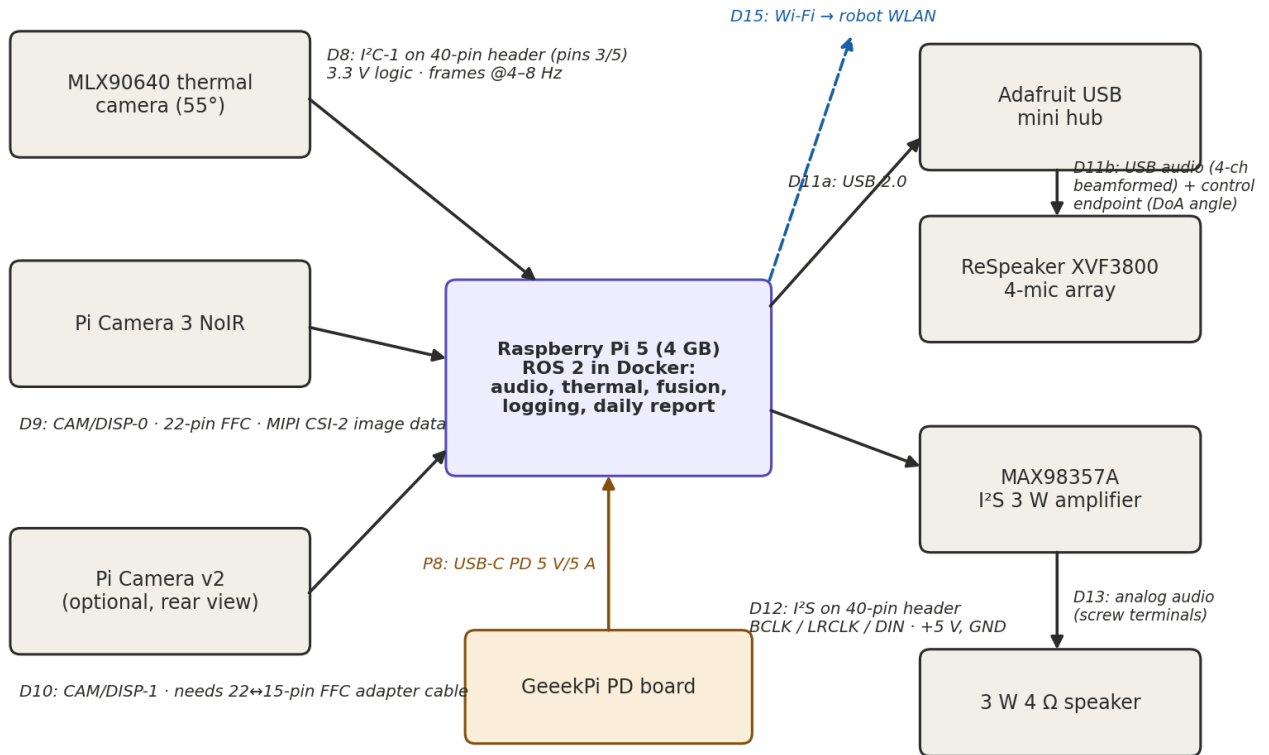
8 · Raspberry Pi 5 peripheral connections

The Pi hosts every sensor that isn't in the navigation loop, plus the speaker, the event log, and the report server.

Table 8-1 · Pi 5 connections

ID	From (connector)	To (connector)	Interface	Signal carried	Electrical / cable notes
D8	MLX90640 breakout (VIN/GND/SDA/SCL)	Pi 40-pin header: 3V3 (pin 1), SDA1 (pin 3), SCL1 (pin 5), GND (pin 6)	I ² C, 3.3 V logic	32×24 thermal frames at 4–8 Hz	Set I ² C baud ≥400 kHz in config for higher frame rates
D9	Pi Camera 3 NoIR FFC	Pi CAM/DISP-0 connector	22-pin mini FFC, MIPI CSI-2	Image data + camera control	Contacts face per the connector marking; night use needs the IR illuminator (P11)
D10	Pi Camera v2 FFC (optional)	Pi CAM/DISP-1 connector	MIPI CSI-2	Image data (spare / rear view)	Requires 22↔15-pin FFC adapter cable
D11	ReSpeaker XVF3800 USB	Adafruit mini hub → Pi USB-A	USB 2.0 (USB audio class + control endpoint)	4-channel beamformed audio; DoA angle via control interface	Bus-powered; keep the array's top face unobstructed
D12	MAX98357A BCLK/LRCLK/DIN/GND	Pi 40-pin header I ² S: GPIO18 (pin 12), GPIO19 (pin 35), GPIO21 (pin 40), GND	I ² S digital audio	Mono playback stream	Amp VIN from the 5 V accessory rail (P11); grounds common at the GND strip
D13	MAX98357A speaker terminals	3 W 4 Ω enclosed speaker	Screw terminals, 2-wire	Analog audio, ≤3 W	Face the speaker forward
D15	Pi Wi-Fi	GL-SFT1200 router	Wi-Fi 802.11n 2.4 GHz	DDS topics; HTTP report on :8080	Static lease (e.g. 192.168.8.11)
D16	(optional) IR LED module gate	Base-controller GPIO via logic-level MOSFET	5 V switched	Illuminator on/off	LED power from 5 V rail (P11)
P8	(power — see Table 5-1)	USB-C connector			

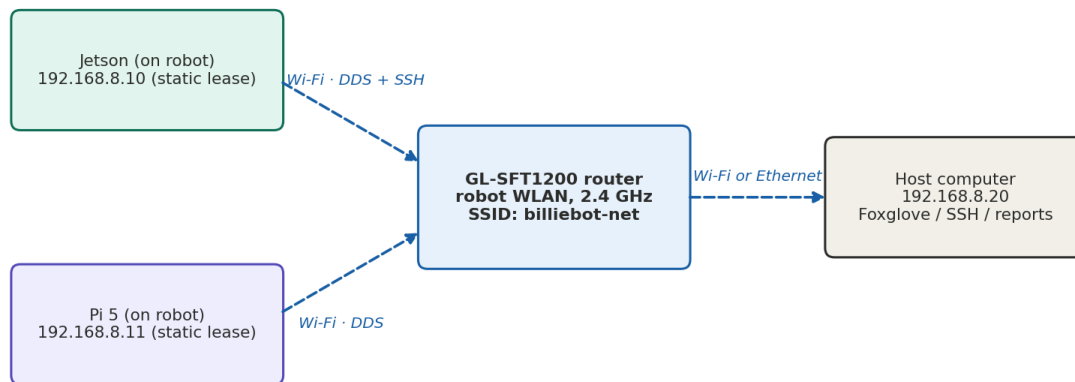
Figure 4 — Raspberry Pi 5 peripheral connections (P8, D8-D13, D15)



9 · Network setup

Configure the GL-SFT1200 as a dedicated robot WLAN and place it centrally in the apartment — it stays off the robot. Use 2.4 GHz (better wall penetration), assign static DHCP leases (Jetson .10, Pi .11, host .20), share one ROS_DOMAIN_ID across all machines, and install the package's CycloneDDS config with a static peers list on each machine (Wi-Fi multicast discovery is unreliable). Verify with a cross-machine **ros2 topic pub / echo** in both directions.

Figure 5 — Network layout (router stays in the apartment, not on the robot)



All three machines share ROS_DOMAIN_ID and a CycloneDDS config with a static peers list (Wi-Fi multicast discovery is unreliable).

10 · Mechanical assembly order

1. Lower deck: motors + wheels + caster; battery centered and low on a velcro strap (it comes off to charge); terminal strips + fuses; L298N (give it airflow — it runs warm); Arduino; BNO055 near the rotation center; GS2678 + GeekPi boards. Route motor power and signal looms on opposite sides of the deck.
2. Push-test: the rolling chassis tracks straight by hand; the caster doesn't shimmy.
3. Upper deck: RPLidar topmost with 360° clearance (nothing in its scan plane — check zip ties); OAK-D forward at 0.2-0.3 m height, tilted 10-15° down (your subject is a 0.15-0.25 m-tall dachshund); thermal + NoIR co-located beside it; ReSpeaker flat on top, unobstructed; Jetson and Pi with ventilation; speaker forward; master switch and fuse block reachable.
4. Dress the harness per Sections 5-8; label every wire with its connection ID.
5. Record the as-built geometry: wheel-center separation, loaded wheel diameter, and each sensor's x/y/z offset + tilt from base center. Enter these in the software package's URDF / MEASURE_ME.md.
6. Weigh the robot (sanity check for motor loading and endurance estimates).

11 · Staged test plan

Stage 1 — Bench power verification (connections P1-P14)

1. Bench PSU at 12.0 V, 0.5 A limit, all load ends disconnected: bus reads 12.0 V; leakage < 50 mA.
2. Set GS2678 to 5.10 V unloaded; confirm it holds 5 V with a brief ~1 A dummy load.
3. Confirm GeekPi PD output presents valid 5 V USB-C.
4. Divider node (P13) reads ≈ 2.98 V at 12.0 V in.
5. Repeat the voltage survey once, briefly, on the LiPo (expect ~11.5-12.6 V depending on charge).

CHECKPOINT: All rails within spec on PSU and battery; every measurement logged.

Stage 2 — Base electronics bench test (D3-D7, P5, P13)

1. Flash the reference firmware; confirm the telemetry frame (ticks, IMU, battery) streams over D3 and matches what `diff_drive_base.py` expects.
2. **Encoders:** hand-rotate each wheel one turn both directions; log ticks/rev; L and R match within a few ticks and change sign with direction.
3. **Motors (wheels-up, PSU limit 2 A):** low PWM turns both wheels; a forward command drives both wheels forward (fix polarity at D6 now); full speed ≈ 130 rpm no-load.
4. **PID:** a 0.1 m/s setpoint settles within ~ 1 s without oscillation.
5. **IMU:** complete the BNO055 calibration routine; yaw tracks hand rotation and returns to start within $\sim 2^\circ$.
6. **Battery ADC:** sweep PSU 10.0 $\rightarrow$ 12.6 V; reported voltage tracks within 0.1 V.
7. **Watchdog (pre-TC-PLT-5):** with wheels spinning, unplug D3; motors stop within 500 ms. If the firmware lacks this, add it before anything else.

CHECKPOINT: All seven items pass; the drive electronics are now a trusted unit.

Stage 3 — Compute and network bringup

1. Jetson: install the NVMe (thermal pads on), flash JetPack 6, set the 15 W power model, install ROS 2 Humble + Nav2 + slam_toolbox + depthai-ros.
2. Pi 5: flash 64-bit OS, attach the hub, build/pull the Humble Docker container.
3. Configure the router per Section 9; SSH keys everywhere.
4. Run the software package's mock ladder (`run_all_mock_tests.sh`) so the code is known-good before hardware.

CHECKPOINT: Cross-machine topic echo works both directions with < 1 s discovery; mock suite green.

Stage 4 — Assembled smoke test

1. Assemble per Section 10; connect everything.
2. Wheels up, PSU at 12 V with a 3 A limit: power on — Jetson and Pi boot; no rail sags; log idle draw (expect ~ 1.5 – 2.5 A at 12 V).
3. Repeat once on the LiPo. After 10 min, touch-test the L298N and regulators: warm is fine, untouchable is not.

CHECKPOINT: Clean boot on battery; thermal survey acceptable; idle current logged.

Stage 5 — First drive (launch rungs 02-03)

1. Wheels-up: rung 02_base — verify `/odom`, `/tf`, `/imu/data`, `/battery_state` (verify script 02).
2. **Odometry calibration** (floor, clear area, hand near the switch): teleop 2.0 m along a tape line — `/odom` reports 2.0 m $\pm 5\%$ (tune ticks/rev, wheel radius). Rotate 360° in place — yaw error $< 10^\circ$ (tune wheel separation). Iterate.
3. Rung 03 (EKF): drive a square; the filtered track closes visibly better than raw odom.
4. Re-run the watchdog on the full robot (wheels up): kill the base node mid-drive — motors stop < 500 ms. File TC-PLT-5.

CHECKPOINT: 15 min of smooth teleop apartment-wide; no USB resets or brownouts in `dmesg`; TC-PLT-5 passed.

Stage 6 — Lidar, mapping, localization (rungs 01, 04, 05)

1. Rung 01: `/scan` ≥ 5 Hz; the scan matches the room in RViz. Smearing during rotation = loose mount or wrong lidar TF yaw.

2. Rung 04: map every room at < 0.2 m/s, closing loops through multiple doorways; save the map.
3. Apartment housekeeping: tuck floor cables; add keepout zones for anything the robot genuinely can't handle (rug edges, chair-leg forests).
4. Rung 05: AMCL on the saved map; particle cloud collapses after a wiggle. Run TC-NAV-2 against 3–4 taped reference points.

CHECKPOINT: Complete saved map; TC-NAV-2 ≤ 0.15 m error.

Stage 7 — Autonomous navigation (rung 06)

1. Set the Nav2 footprint from measured chassis dimensions; 0.3 m/s max; load map + keepouts.
2. Goals: same room → cross-room → through doorways (this is where the tuning time goes).
3. Run TC-NAV-3 (walk into its path), TC-NAV-4 (block it; recovers then SAFE after 3 fails), define the patrol YAML, run TC-NAV-6 (10 laps, zero touches).
4. Run TC-PLT-2 (mock descending voltage → SAFE); note real per-lap battery drain.

CHECKPOINT: TC-NAV-3/4/6 and TC-PLT-2 pass; unattended patrol while you watch from the couch.

Stage 8 — Perception sensors one at a time (rungs 07–11)

1. **OAK-D (07/08):** RGB/depth verified; tune the dog-detection confidence on real Billie images (a dachshund is a low, long silhouette — expect to lower the threshold and re-check false positives on shoes and draft stoppers). Run TC-PER-1 and TC-PER-2 at taped 1/2/3 m marks; confirm /dog/pose_map lands in the right room.
2. **Thermal (09):** warm target ≤ 1.5 m in darkness → blob; cold decoy → none. Run TC-PER-3.
3. **NoIR (10):** low-light demo, TC-PER-5. If it fails without the IR illuminator, wire D16/P11 now or formally accept thermal-only night coverage.
4. **Audio (11):** record 20–50 real Billie barks/whines plus negatives (doorbell, TV, your voice, vacuum) into the test assets; tune threshold; run TC-PER-4 including DoA bearings ($\pm 15^\circ$).
5. **Speaker:** quiet chime plays; the rate limiter blocks rapid repeats.

CHECKPOINT: TC-PER-1...5 filed (waivers documented, e.g. PER-5 pending illuminator).

Stage 9 — Cognition and mission (rungs 12–13)

1. Rung 12: inject synthetic states; run TC-STL-2 and TC-STL-4.
2. Record ~1 h of Billie living normally, hand-annotate, replay for TC-STL-1; tune sleep/rest/active thresholds (a lying dachshund is a very wide, flat bounding box).
3. Rung 13 with mock inputs: PATROL → INVESTIGATE on injected bark (TC-FND-2); TRACK_OBSERVE on injected detection; standoff never < 1.0 m (TC-FND-3).
4. Seed a synthetic day; run TC-RPT-1 and TC-RPT-2 (open the report from the host browser).

CHECKPOINT: The whole pipeline works with a fake dog.

Stage 10 — Live trials with Billie (short, calm, positive)

1. Session A: robot powered but stationary during normal life, treats scattered near it; two or three 10-min sessions across days.
2. Session B: slow teleop, never toward Billie, with treats. Stop early on stress signals (tail tuck, retreat, fixed staring).
3. Session C: first autonomous patrols with Billie present, you supervising; run TC-NAV-5 live (near-dog speed cap visibly working).
4. Run TC-FND-4: 10 find-the-dog trials, Billie in different rooms; between trials, check event-log rows and snapshots.

CHECKPOINT: TC-FND-4 $\geq 8/10$ within 10 min; Billie relaxed around the robot. Her comfort is a hard requirement — the future no-stress learning depends on it.

Stage 11 — MVP acceptance (rung 14)

1. Split-host bringup on boot (systemd); scheduled patrols.
2. TC-PLT-1 endurance: one full battery, 12.6 V to SAFE trigger, ≥ 60 min while patrolling; save the telemetry curve.
3. 24 h soak with Billie: patrols on schedule, events logged, the 23:30 report generated. Compare the report against your own observation of her day.
4. Close the verification matrix: every TC has a result file; every requirement has a pass or a documented waiver.

CHECKPOINT: MVP complete. The event database now begins accumulating the (context, action = OBSERVE, outcome) baseline the v2 contextual bandit will learn from.

12 · Troubleshooting quick reference

Symptom	First suspects
Jetson reboots when motors start	Voltage sag: check the P14 bulk cap, wire gauge/length on P6, shared ground path; reduce motor acceleration
One wheel runs backward	Swap that motor's OUT pair at the L298N (D6)
Odometry drifts in turns	Wheel-separation calibration (Stage 5.2); tire slip on rugs — slow the rotation speed
Map smears / doubled walls	Loose lidar mount, wrong lidar TF, mapping too fast
OAK-D disconnects randomly	USB power (D2): powered hub or Y-cable from the 5 V rail; try a better cable
AMCL lost after startup	Wrong initial pose; raise particle count; relocalize near a feature-rich corner
No cross-machine topics	CycloneDDS peers list / firewall; confirm identical ROS_DOMAIN_ID
False dog detections	Raise the confidence threshold; add hard-negative images (shoes, bags) to the tuning set
Bark classifier misses Billie	The recorded-bark set is too small; collect more samples and re-tune
Thermal frames garbled	I ² C (D8) baud too low or wiring too long; shorten leads, check pull-ups
No speaker output	I ² S overlay not enabled on the Pi; check D12 pin mapping and P11 amp power
L298N very hot	Normal-ish (it is lossy) — heatsink/airflow now; plan a modern driver swap in v2

Appendix A · Master connection checklist

Tick each connection off as it is made and verified. Full details in the tables of Sections 5–8.

✓	ID	Connection	Interface	Signal
☐	P1	Battery (+) → 20 A main fuse	XT60 / solder	Raw 12 V
☐	P2	Main fuse → master switch	Spade	Switched 12 V
☐	P3	Switch → +12 V bus strip	Ring terminal	Bus supply
☐	P4	Battery (–) → GND bus strip	Ring terminal	Star ground
☐	P5	Bus [10 A] → L298N VMOT/GND	Screw terminals	Motor power
☐	P6	Bus [10 A] → Jetson barrel 5.5×2.5 mm	Barrel plug	Compute power 12 V
☐	P7	Bus [5 A] → GeekPi PD VIN	Screw terminals	PD input
☐	P8	GeekPi USB-C → Pi 5 USB-C	USB-C	5 V/5 A PD
☐	P9	Bus [5 A] → GS2678 IN	Screw terminals	Buck input
☐	P10	GS2678 OUT → 5 V accessory strip	Screw terminals	5.1 V rail
☐	P11	5 V rail → amp VIN, IR LED	Dupont / solder	Accessory 5 V
☐	P12	Bus → Pololu D24V90F5 (output taped)	Solder pads	Reserved rail
☐	P13	Bus → divider → base-controller ADC	22 AWG	0–3.1 V battery sense
☐	P14	≥1000 µF cap across Jetson feed	Solder	Ride-through
☐	D1	RPLidar (via UART-USB adapter) → Jetson	USB 2.0	Lidar scan serial
☐	D2	OAK-D Lite → Jetson blue port	USB 3.0	RGB/depth + detections
☐	D3	Arduino Nano → Jetson	USB 2.0	Base serial 115200 + Nano power
☐	D4	Encoders L/R → base controller	Dupont	Quadrature A/B, 5 V
☐	D5	Base controller → L298N IN/EN	Dupont	PWM + DIR logic
☐	D6	L298N OUT pairs → motors L/R	Screw terminals	12 V chopped drive
☐	D7	BNO055 → base controller	I ² C, 5 V	Orientation @100 Hz
☐	D8	MLX90640 → Pi header pins 1/3/5/6	I ² C, 3.3 V	Thermal frames
☐	D9	Pi Cam 3 NoIR → Pi CAM/DISP-0	22-pin FFC, CSI-2	Image data
☐	D10	Pi Cam v2 → Pi CAM/DISP-1 (optional)	22↔15-pin FFC, CSI-2	Image data
☐	D11	ReSpeaker → hub → Pi	USB 2.0 audio class	4-ch audio + DoA

✓	ID	Connection	Interface	Signal
☐	D12	MAX98357A → Pi I ² S pins 12/35/40	I ² S	Digital audio out
☐	D13	Amp → 3 W 4 Ω speaker	Screw terminals	Analog audio
☐	D14	Jetson ↔ router	Wi-Fi 2.4 GHz	DDS + SSH
☐	D15	Pi ↔ router	Wi-Fi 2.4 GHz	DDS + HTTP :8080
☐	D16	IR LED gate ← base GPIO via MOSFET (optional)	5 V switched	Illuminator control